

CYBERSECURITY AI ASSISTANT ANALYSIS

Questions

1. Explain what happens to the `worker_responses` list as each expert node completes its task. Why is `operator.add` essential for this design pattern?

Solution

As each expert node (such as the Researcher or Editor) completes its task, its output is appended to the `worker_response` list. `Operator.add` is essential for the design pattern because it allows the nodes to operate parallelly so that it doesn't override the previous results. The list **acts as a shared memory state that grows as the conversation progresses.**

`Operator` is essential for the **Map-Reduce** and **Parallel** design patterns because it ensures that new data is added to the list rather than overwriting the previous expert's work. Without `operator.add`, the "Editor" node would simply replace the "Researcher's" findings with its own, leaving the final Manager with an incomplete context

2. Which model was the fastest on your hardware? Did the DeepSeek-R1 reasoning model provide a deeper analysis than the standard Llama model?

Solution

Fastest Model: In terms of Expert nodes the fastest model on the hardware was the Researcher node (often using a smaller or more specialized model) completing its task in as little as **3.65s**.

In respect to quantized models the **Q2 quantized** version of the models was significantly faster on the hardware. Specifically, the Manager's latency dropped from **130.52s** in the Q8 trial to **65.96s** in the Q2 trial.

DeepSeek-R1 vs. Llama: The DeepSeek-R1 reasoning model provided a notably deeper analysis. In Trial B, it acted as the "Threat Hunter," identifying specific communication flows between IP addresses and security protocols, whereas the standard Llama summary was more generalized.

3. Describe the "Model Swap" latency or memory pressure you observed when Ollama switched between different models in your graph.

Solution

When switching between models (e.g., DeepSeek-R1 for research and Llama for management), significant resource strain was noted:

- Latency Spikes: The Manager node experienced high latency (up to 130.52s) partly due to the overhead of coordinating different models.
- Memory Pressure: In high-precision tests (Q8), the memory pressure was severe enough that the system often failed to complete the final response

4. Identify at least one instance where a smaller or more quantized model provided an inaccurate fact or hallucination. How did your Human-in-the-Loop feedback or the Manager's re-routing address this error?

Solution

A clear instance of hallucination occurred with the 2-bit quantized (Q2) version of Llama 3:8b.

- The Error: The Q2 model "lost the logical thread," hallucinating the benefits of a Null Scan and failing to provide coherent drawbacks.
- Human-in-the-Loop (HITL) Fix: In Trial B, when the model failed to mention packets 5-10, the user provided direct feedback.
- Manager Re-routing: The Manager node then re-routed the task back to the Researcher and Editor nodes to incorporate the missing data, resulting in a revised, comprehensive response

5. Tabulate the response quality and latency of each expert across three different topics: Technical, Ethical, and Creative.

Solution

Topic	Researcher(Deepseek)	Editor(Qwen)	Latency(sec)
Technical	9/10 (Captured detailed IP flows)	9/10 (Approved complex network summary)	191.1sec
Ethical	7/10 (Structured analysis of Null Scans)	8/10 (Refined pros/cons list)	130.52sec
Creative	4/10 (Fast, but omitted packets 5-10 in	5/10 (Initially missed the	28.48sec

	summary)	hallucinated data)	
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